CRIMEVISIONAI

Proposed Solution

Folder 3 | Solution proposal | Prepared 23 October 2026

Project: CrimeVisionAI — Multimodal Forensic Evidence Analysis and Investigation Support System

Document status: Project-specific draft based on the implementation information currently available. Verify placeholders and run-dependent results before formal submission.

1. Executive summary

CrimeVisionAI is a browser-based multimodal forensic evidence analysis and investigation-support prototype. It organizes case records and uploaded evidence, runs AI-assisted analysis for audio, images and video, produces candidate events and timeline-oriented outputs, and provides case-level investigator Q&A; over available analysis.

2. Objectives

• Reduce repeated manual switching between evidence types. • Provide first-pass transcripts, object detections, captions and scene cues. • Organize candidate events chronologically where timestamps are available. • Make evidence and analysis accessible through a case-centric workflow. • Keep a human reviewer responsible for interpretation and decisions.

3. Proposed workflow

Step	Action	System output
1	Create/select a case	Case metadata and case context.
2	Upload supported evidence	Stored file and evidence metadata.
3	Choose modality analysis	Audio, image or video analysis pipeline.
4	Inspect model results	Transcript, timestamps, detections, scene labels, captions and cautious interpretation.
5	Review timeline	Candidate events ordered by available timestamps.
6	Ask investigator questions	LLM-assisted answer based on available case analysis; verify against evidence.
7	Document findings	Manual review and notes; formal PDF report export remains unconfirmed.

4. Core implementation

Backend: Flask 3.x app factory. Frontend: Jinja2 templates with HTML, CSS and vanilla JavaScript. Data: SQLite through raw sqlite3, with cases, evidence and analysis_results tables. File storage: local evidence/uploads and case-scoped upload paths. Models/integrations: Whisper base, RT-DETR rtdetr-l.pt, CLIP ViT-B/32, BLIP captioning and Groq openai/gpt-oss-20b.

5. Scope and limitations

Current baseline supports audio formats WAV/MP3/M4A/FLAC/OGG/AAC/WMA and video MP4/MOV/AVI/MKV/WEBM. Audio is chunked into 30-second segments; video is sampled at 1 FPS and capped at 180 seconds / 180 frames. No speaker diarization is reported. Q&A; has a known grounding/completeness defect. PDF report generation and 3D visualization are not confirmed complete.

6. Expected outcomes

The expected outcome is a demonstrable end-to-end prototype for organizing and reviewing multimodal evidence. Numerical claims about accuracy, speed, investigator time saved or deployment capacity are intentionally excluded until measured.